

Solution Requirements

Date	1 July 2026
Team ID	XXXXXX
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users and administrators should securely log in to access the flood prediction system and dashboard.	High
2	Authorization levels	Different access permissions should be provided for admin, researchers, and end users.	High
3	External interfaces	System should connect with weather APIs, rainfall datasets, river-level monitoring systems, and cloud database.	High

4	Transactions processing	System should collect real-time environmental data and process it for flood prediction analysis.	High
5	Reporting	System should generate flood risk reports, prediction results, alerts, and visualization dashboards.	Medium
6	Business rules, etc.	Prediction model should analyze rainfall, water level, humidity, and historical flood data before generating output.	High
7	Compliance to laws or regulations	System should securely handle environmental and public safety data according to data privacy standards.	Medium
8	<i>Other</i>	System should send early warning notifications when flood risk crosses the danger threshold.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	System should analyze incoming weather and water-level data quickly for prediction.	Prediction generated within 2 seconds

2	Scalability	System should support monitoring data from multiple regions simultaneously.	Supports 50,000+ data records
3	Security & Data Privacy	Collected environmental and user data must be securely stored and protected.	AES-256 secure encryption
4	Reliability & Availability	System should remain active continuously during critical weather conditions.	99.9% uptime monthly
5	Usability & Accessibility	Dashboard should be simple, user-friendly, and accessible on mobile and desktop devices.	Responsive interface with easy navigation
6	<i>Other</i>	System should allow future machine learning model improvements and dataset expansion.	System should allow future machine learning model improvements and dataset expansion.